CHAUDHARY CHARAN SINGH UNIVERSITY, MEERUT • BCA-508

VAULTY

Personal Subscription & Burn-Rate Intelligence Web Application

A full-stack, client-first financial intelligence dashboard designed to eliminate subscription fatigue, terminate zombie expenses, and defend against predatory trial traps.

SUBMITTED BY:

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INTRODUCTION & PROBLEM STATEMENT

The Real-World Subscription Crisis

Macro Market Challenges

- **Shift to Recurring Billing:** Entertainment (Netflix, Spotify), Cloud (Google One), AI (Claude Pro, ChatGPT), and Gym platforms charge recurring fees rather than one-time costs.
- **Severe Spend Blindness:** Studies indicate 84% of consumers significantly underestimate monthly recurring outflows by 2x to 3x.
- **Silent Auto-Debit Leaks:** Fractional charges (₹149, ₹299, ₹649) silently compound into an annual bleed of ₹45,000+ per student/developer.

Identified Flaws in Existing Solutions

- **Zombie Subscriptions:** Forgotten memberships unused for >30 days continue charging accounts perpetually.
- **Dark Pattern Trial Traps:** Free trials require payment card details and quietly auto-convert without user reminder.
- **Roommate Split Friction:** Shared subscriptions (Netflix 4K, Spotify Duo) lack automated payment tracking and UPI links.
- **Data Privacy Risks:** Banking aggregation apps demand sensitive OTPs and full net-banking credentials.

PROJECT OBJECTIVES

Core Goals & System Capabilities

1. Unified Spend Ledger

- **One-Pane Dashboard:** Aggregate all recurring digital commitments on a single intuitive Bento screen.
- **30-Second Quick Presets:** Instant brand presets (Netflix, Spotify, Prime, ChatGPT) to eliminate form-filling friction.
- **Mathematical Normalization:** Automatically convert all billing frequencies to a standard 30-day monthly burn rate.

2. Proactive Defense Engine

- **Bleed Radar:** Automated background scan flagging active subscriptions dormant for ≥ 30 days.
- **Trial Conversion Shield:** Visual countdown for trials ending within 48 hours with 1-click cancellation links.
- **Price Hike Watchdog:** Audits stealth cost adjustments and foreign currency fluctuations across cycles.

3. Enterprise Architecture

- **Zero-Setup Demo Mode:** Works 100% offline via LocalStorage fallback without requiring database connection.
- **Supabase PostgreSQL:** Secure cloud storage enforcing PostgreSQL Row Level Security (RLS) tenant isolation.
- **100% Tested Logic:** Core mathematical engine verified via 7 Vitest unit tests.

SYSTEM ARCHITECTURE

3-Tier Layered Structural Design

Tier 1: Presentation Layer

- **Next.js 14 App Router:** React Server Components for optimal page speeds and zero-waterfall renders.
- **Dark Glassmorphic UI:** Tailored #090A0F theme, backdrop blur filters, and vibrant neon status indicators.
- **Interactive Recharts:** Declarative SVG Burn Rate Flow area trajectory and category donut breakdown.

Tier 2: Business & Math Layer

- **Pure Math Engine:** `lib/calc.ts` decoupled from UI renders for deterministic, sub-millisecond calculation.
- **Zod Schema Validation:** Strict type verification preventing runtime edge-case data corruption.
- **Command Palette:** Instant global keyboard navigation powered by `cmdk` (⌘K / Ctrl+K).

Tier 3: Persistence Layer

- **Repository Pattern:** Decouples UI from underlying storage mechanism via unified interface.
- **Offline LocalStorage:** Immediate zero-latency demo mode with instant persistence.
- **PostgreSQL + RLS:** Cloud persistence with strict tenant isolation enforced at the database engine level.

CORE ALGORITHMS

Mathematical Calculation Engine (lib/calc.ts)

Core Computational Formulas

- **1. Billing Cycle Normalization:**
  `C_monthly = Cost × (365 / (12 × D_cycle))`
  Normalizes Weekly (×4.345), Monthly (×1), Quarterly (/3), and Annual (/12) to exact 30-day burn rate.
- **2. Monthly Bleed Summation:**
  `Total Bleed = Σ C_monthly` for all records with `status == 'active'` (paused and cancelled subs excluded).
- **3. Zombie Inactivity Boundary:**
  `isZombie = (Current_Date − Last_Used_Date) ≥ 30 days`.
- **4. Trial Trap Urgency Horizon:**
  Flagged as urgent when `(Renewal_Date − Current_Date) ≤ 48 hours`.

Automated Vitest Verification (7/7 Passed)

- **Test 1: Cycle Normalization:** Verified mathematical precision across all 4 billing cadences.
- **Test 2: Status Isolation:** Confirmed paused and cancelled subscriptions contribute exactly ₹0 to burn rate.
- **Test 3: 30-Day Boundary Precision:** Validated strict boundary (29 days = active, 30 days = zombie).
- **Test 4: Leap Year & Roll-Forward:** Verified accurate renewal date advancement past month-ends.
- **Test 5: Zero Division Safety:** Handles zero-value deltas without NaN errors.
- **Test 6: Trial Window Isolation:** Accurately isolates trials expiring within target window.
- **Test 7: Seed Calibration:** Baseline dataset reliably produces exactly ₹3,849/month.

USER INTERFACE

Key Feature Modules & Walkthrough

1. Bento Dashboard

- **Wallet Bleed Card:** Displays ₹3,849/mo hero metric with 6-month historical sparkline.
- **Zombie Sub Card:** Highlights 3 idle subscriptions with instant ₹899/mo savings review.
- **Trial Trap Radar:** 48-hr warning for Prime trial with 1-click cancel deep-link.
- **Burn Rate Flow:** 12-month area chart forecasting year-end spend trajectory.

2. Sub Vault

- **Quick Brand Presets:** Instant brand presets for Netflix, Spotify, Prime, Claude, ChatGPT, Cult.fit.
- **Multi-Select Filtering:** Real-time filter pills across Streaming, AI Tools, Cloud, Gaming, Zombies, Trials.
- **Instant Status Toggle:** Flip status between Active and Paused in one click.
- **CSV Export:** One-click complete ledger export for offline auditing.

3. Bleed Radar

- **Zombie Terminator:** Identifies dormant services and calculates recoverable monthly cash.
- **Price Hike Watchdog:** Flags stealth price jumps (+₹20 Spotify Duo hike).
- **Savings Simulator:** Interactive projection of 1-year and 3-year bank balance gains.
- **Notification Center:** Priority-ranked in-app notifications for imminent debits.

4. Smart Split

- **Shared Cost Matrix:** Splits multi-user plans (e.g., Netflix 4K with roommates).
- **True Net Bleed:** Computes actual personal liability after split reimbursements.
- **WhatsApp Reminders:** Pre-formats polite WhatsApp reminder messages.
- **Instant UPI Links:** Generates 1-click UPI payment deep-links for instant settlement.

EXPERIMENTAL RESULTS

Calibrated Baseline (Student Persona: Arjun)

Active Subscriptions Ledger (₹3,849/mo)

- **Netflix 4K (Streaming):** ₹649.00 / month — Renews in 3 days
- **Cult.fit Gym (Gym/Health):** ₹899.00 / month — Unused for 38 days (ZOMBIE)
- **Claude Pro (AI Tools):** ₹1,700.00 / month ($20 USD normalized) — Active daily
- **Spotify Duo (Streaming):** ₹149.00 / month — Renews in 7 days
- **Google One 100GB (Cloud):** ₹130.00 / month — Continuous backup
- **Xbox Game Pass PC (Gaming):** ₹322.00 / mo (₹967/quarter) — Unused 42 days (ZOMBIE)

Recoverable Savings & Defensive Alerts

- **3 Zombie Subscriptions Identified:** Cult.fit Gym (₹899) + Xbox Game Pass (₹322) + iCloud (₹75) = ₹899/month recoverable.
- **Annualized Cash Retained:** Eliminating the 3 detected zombies saves ₹10,788 per calendar year.
- **Urgent Free Trial Defended:** Amazon Prime trial ends in 48 hours. Direct cancel deep-link surfaced before credit card charge of ₹299.
- **Smart Split Reimbursement:** Splitting Netflix 4K with 3 roommates recovers ₹486.75/month back to user's UPI.

TECHNOLOGY ARCHITECTURE

Software, Libraries & Development Tools

Frontend Framework & UI

- **Next.js 14 (App Router):** React Server Components for optimal SEO and client rendering speed.
- **React 18:** Concurrent mode and optimistic UI updates for instant interactions.
- **Tailwind CSS 3.4:** Utility-first CSS variables powering dark glassmorphism.
- **Recharts:** Declarative SVG area charts, sparklines, and donut graphs.
- **Lucide React:** Ultra-lightweight scalable vector UI icon library.

Backend, Database & Math

- **TypeScript 5:** Strict static typing across all schemas, props, and calculation logic.
- **Supabase Cloud:** Managed PostgreSQL relational database with Row Level Security.
- **date-fns Engine:** Immutable temporal arithmetic for leap-year-safe renewal calculation.
- **Zod Validation:** Schema declaration and runtime input validation.

Testing, Build & DevOps

- **Vitest:** Sub-millisecond unit test suite testing calculation theorems.
- **ESLint & PostCSS:** Code quality inspection and CSS optimization.
- **Headless Chrome Engine:** High-fidelity PDF compilation with native link annotations.
- **Vercel Edge CI/CD:** Automated build verification and edge distribution.

CONCLUSION & VERIFICATION

Project Summary & Presentation Deliverables

Project Summary & Achievements

- **100% Curriculum Compliance:** Fully satisfies all BCA-508 minor project guidelines for Chaudhary Charan Singh University.
- **Sub-Millisecond Engine:** Zero-latency calculations decoupled from visual presentation.
- **Production Tested:** 7/7 Vitest unit tests pass; all 13 Next.js routes compile statically with 0 errors.
- **Dual Persistence:** Instant zero-setup demo mode + secure PostgreSQL cloud syncing.

Mandatory Materials Ready for Tomorrow

- **1. Project Report PDF (Clickable):** `VAULTY_Minor_Project_Report.pdf` (33 Pages with 152 active hyperlinks).
- **2. Project Source Code Bundle:** `VAULTY_Source_Code.zip` packaged and ready for pen-drive submission.
- **3. Mandatory Presentation PPT:** `VAULTY_Minor_Project_Presentation.pptx` (and this PDF slide deck).
- **4. Live Application Instance:** Running locally via `start.command` on port 3000.

THANK YOU!

Questions & Live Demonstration

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